

# 02LQD Specification and Simulation of Digital Systems

2018/07/02

## 1. Combinational circuit design (5 points)

8 4 -2 -1 is a self-complementary, biased and redundant code. Each decimal digit is expressed on 4 bits. The following table matches BCD and 8-4-2-1:

| BCD  | 8 4 -2 -1 |
|------|-----------|
| 0000 | 0000      |
| 0001 | 0111      |
| 0010 | 0110      |
| 0011 | 0101      |
| 0100 | 0100      |
| 0101 | 1011      |
| 0110 | 1010      |
| 0111 | 1001      |
| 1000 | 1000      |
| 1001 | 1111      |

Design in VHDL a BCD to 8-4-2-1 transcoder implementing the above table according to the following description styles:

1. dataflow using conditional signal assignments (WHEN ELSE statements)
2. dataflow using selected signal assignments (WITH SELECT WHEN statements)

## 2. FSM-D design (25 points)

The factorial of a positive integer is recursively defined as follows:

$$\begin{aligned} \text{fact}(n) &= n * \text{fact}(n-1) & n > 1 \\ \text{fact}(0) &= 1 \end{aligned}$$

Design a FSM-D to **iteratively** compute the factorial of a positive integer  $n$  expressed on 3 bits. Note that, being  $\text{fact}(7)=5040$ , the output `data_out` is on 13 bits. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. The output `data_out` makes sense only when `ready` is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (10 points)
2. Design a FSM-D (15 points):
  - create 2 processes for behavioural datapath description from the HLSM state diagram
  - create 2 processes for behavioural controller description from the HLSM state diagram
  - create a process for behavioral output description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
    - running VHDL code and testbench
    - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
- no later than Thursday July 5 1pm**
- remember: no upload  $\Rightarrow$  no correction  $\Rightarrow$  no oral exam
  - once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to [paolo.camurati@polito.it](mailto:paolo.camurati@polito.it) to agree on a date for the oral exam.